

Image Plagiarism Detection with Equivariant Encoders

Denis Gudkov

Consultant: Daniil Dorin

Advisor: Andrey Hrabovoy

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Intelligent Systems Department, MIPT

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Goals

Plagiarists modify images by rotating, reflecting, shifting, and re-compressing them. A good detector must stay invariant to these transformations.

Hypothesis

Embedding exact G -invariance ($G \in \{D_4, SE(2)\}$) into a ViT encoder reduces FPR and raises Recall on geometric transformations from G , compared to a non-equivariant ViT-Tiny/16 trained with geometric augmentation.

Problem

Standard ViTs have no geometric invariance; augmentation only covers group orbits stochastically and gives no guarantee on unseen transforms.

Solution

- 1) Build a siamese detector with encoders spanning $D_4, SE(2)$.
- 2) Train each without geometric augmentations.
- 3) Evaluate FPR and Recall on balanced DomainNet.

Central Idea - Equivariant Encoders

Let G act on the image space $\mathcal{X}$ and let $\rho: G \rightarrow \text{GL}(d)$ be a linear representation of G on $\mathbb{R}^d$. An encoder $\phi: \mathcal{X} \rightarrow \mathbb{R}^d$ is G -equivariant if

$$\phi(g \cdot x) = \rho(g) \phi(x) \quad \forall x \in \mathcal{X}, \forall g \in G.$$

- 1) Inputs: pair (x_1, x_2) ; label $y \in \{0, 1\}$.
- 2) Encoder: ϕ_θ , shared, G -invariant.
- 3) Head: symmetric fusion h_ψ , MLP.
- 4) Loss: weighted BCE + $\lambda \mathcal{L}_{\text{contr}}^{L^2}$.
- 5) Groups: D_4 (Octic), $SE(2)$ (Harmformer).

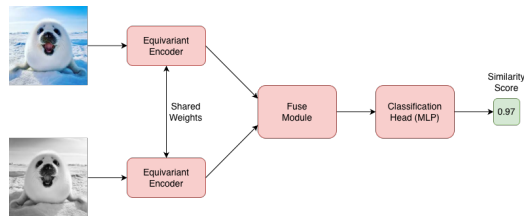


Figure: Siamese pipeline: shared equivariant encoder, fuse module, classification head $\rightarrow$ similarity score.

Related Work

- 1) **Bromley et al., 1993** — siamese networks with shared weights and learned pairwise distance.
- 2) **Dosovitskiy et al., 2021** — Vision Transformer (ViT): image patches as tokens, transformer encoder, state-of-the-art on ImageNet at scale; foundation for ViT backbones in vision.
- 3) **Nordström et al., 2025 (Octic ViT)** — D_4 -equivariant ViT; 40% FLOPs reduction.
- 4) **Karella et al., 2024 (Harmformer)** — $SE(2)$ -equivariance via circular harmonic filters.

Problem Statement

Given a pair $(x_1, x_2) \in \mathcal{X} \times \mathcal{X}$, predict $y \in \{0, 1\}$, where $y = 1$ marks x_2 as a plagiarized derivative of x_1 .

The dataset is $\mathcal{D} = \{(x_{1,i}, x_{2,i}, y_i)\}_{i=1}^N$ with $x_{\cdot,i} \in \mathcal{X} = [0, 1]^{C \times H \times W}$ and $y_i \in \{0, 1\}$.

Plagiarism detector:

$$\phi: \mathcal{X} \times \mathcal{X} \rightarrow \{0, 1\}$$

Solution

Model. Siamese detector with shared G -invariant ϕ_θ and symmetric head h_ψ .

Let $z_i = \phi_\theta(x_i)$; build pair features

$$\delta = |z_1 - z_2|, \quad p = z_1 \odot z_2 \text{ (element-wise)}, \quad v = [\delta; p] \in \mathbb{R}^{2d},$$

The *fusion* MLP is $w_\psi: \mathbb{R}^{2d} \rightarrow [0, 1]$, $w_\psi(v) = \sigma\left(c^\top \text{ReLU}(Av + b) + d\right)$

The detector output is

$$f_\theta(x_1, x_2) = w_\psi(v) = h_\psi(\phi_\theta(x_1), \phi_\theta(x_2)), \quad \phi_\theta(g \cdot x) = \phi_\theta(x) \quad \forall g \in G.$$

Loss. $\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda \mathcal{L}_{\text{contr}}^{L^2}$, where $z_i = \phi_\theta(x_i)$ and

$$\mathcal{L}_{\text{contr}}^{L^2}(z_1, z_2, y) = y \|z_1 - z_2\|_2^2 + (1-y) \max(0, m - \|z_1 - z_2\|_2)^2.$$

Computational Experiment

Encoders. ViT-Tiny/16 (baseline), OcticViT (D_4), Harmformer ($SE(2)$).

Setup.

- ▶ Training set: COCO 2017 (118K images), cross-batch pairing.
- ▶ Test set: DomainNet, 6 domains, 1 200 pairs per regime (600 pos + 600 neg). Two regimes: *overall* (training-time augs on both branches) and *per-transformation* (photometric off, one geometric transform).
- ▶ Augmentation pipeline (each transform independent, $p = 0.3$):
 - Non-group* (shared): CJ, GBlur, GN, JPEG, WM.
 - Group* (added to ViT-Tiny/16 per comparison):
 - D_4 : R90/180/270, HFlip, VFlip.
 - $SE(2)$: continuous rotation $\alpha \sim \text{Unif}[0, 2\pi)$, translation $\pm 20\%$.
- ▶ Decision threshold $\tau = 0.5$; 20 epochs, AdamW, $B = 128$, $\lambda = 0.05$, $m = 0.5$.

Geometric transformations

Name	Geometric transformation g	Probes
R_{90}	90° rotation	D_4
R_{180}	180° rotation	D_4
R_{270}	270° rotation	D_4
HFlip	horizontal flip	D_4 reflection
VFlip	vertical flip	D_4 reflection
R_{45}	45° rotation	$SE(2) \setminus D_4$
R_{135}	135° rotation	$SE(2) \setminus D_4$
$T_{0.2}$	translation by $\text{Unif}([-0.2, 0.2]^2)$	$SE(2)$ translation

Per-transformation regime: photometric block off, only the corresponding geometric transform on the positive view; 1 200 pairs per transformation.

Results: per-transformation Recall and overall FPR

Model	Overall		Per-transformation Recall %							
	FPR %	Rec %	R_{90}	R_{180}	R_{270}	HFlip	VFlip	R_{45}	R_{135}	$T_{0.2}$
ViT-Tiny (photo only)	7.67	99.83	11.7	16.5	10.7	31.3	21.2	0.5	0.3	1.2
ViT-Tiny (D_4 augs)	3.67	100.0	100.0	100.0	100.0	100.0	100.0	0.3	0.3	1.0
OcticViT	1.17	99.83	100.0	100.0	100.0	100.0	100.0	0.3	0.3	0.5
ViT-Tiny ($SE(2)$ augs)	16.17	95.83	100.0	100.0	100.0	100.0	100.0	31.0	30.7	99.2
Harmformer	12.00	99.83	100.0	100.0	100.0	94.2	94.2	0.7	0.7	2.5

All encoders share non-group augs (CJ, GBlur, GN, JPEG, WM); $\tau = 0.5$.

Results and conclusions

D_4 comparison. OcticViT (no group augs) matches the augmented ViT-Tiny/16 at 100% Recall on every D_4 transformation and cuts overall FPR from 3.67% to 1.17% — architectural invariance fully replaces the D_4 augmentation block and triples precision at fixed τ .

$SE(2)$ comparison. Verdict is mixed. Harmformer beats the augmented baseline on overall FPR (12.00 vs 16.17) and Recall (99.83 vs 95.83), and is exact on the discrete D_4 orbit, but collapses on R_{45}/R_{135} ($< 1\%$) and translation $T_{0.2}$ (2.5%). We attribute the off-axis collapse to interpolation noise at the 32×32 input grid of the released Harmformer code, not to the equivariance principle.

Conclusions.

- 1) For $G = D_4$ the hypothesis H_1 is supported: invariant encoder $\geq$ augmented baseline on both FPR and Recall.
- 2) For $G = SE(2)$ H_1 holds on overall metrics but fails on off- D_4 transformations; resolution-bound, not principle-bound.
- 3) Future work: rerun at higher input resolution; extend the set of transformations to scale and projective ones.